Global Margin Conditioning Amplifies a Gaussian Posterior Mean

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Abstract

We study a Gaussian-process classification prior conditioned to have a fixed absolute margin at every point of the Boolean cube, followed by one noiseless label observation. For the fixed-depth iterated-arcsine kernels arising at the $\gamma = 0$ neural-network scaling, the posterior mean has squared reproducing-kernel Hilbert space norm at least $(1 - o(1))vd \log 2$, where v is the smallest matrix covariance eigenvalue. Consequently, at depth at least three, a dimension-uniform mean-energy estimate of order $n/\tau_{\alpha d}$ is false already at $n = 1$; here $\tau_{\alpha d}$ is the high-degree spectral trace. The proof retains the global-margin conditioning exactly. It identifies an obstruction to transferring an unconditioned Gaussian learning argument, rather than establishing a sample-complexity bound.

1 Model and result

Let $\mathcal{X}_d = \{-1, 1\}^d$, $M = 2^d$, and fix a depth $L \geq 2$. Set

$$A(t) = \frac{2}{\pi} \arcsin t, \qquad F = A^{\circ(L-1)}, \qquad K(x, y) = F\left(\frac{1 + x \cdot y}{d + 1}\right), \qquad \beta = 2^{-(L-1)}.$$

The covariance matrix K has diagonal one. The Walsh characters diagonalize the normalized integral operator K/M; denote its eigenvalue at degree j by λ_j, and write

$$v = \lambda_{\min}(K) > 0, \qquad \tau_r = \sum_{j>r} \binom{d}{j} \lambda_j.$$

For a vector h its squared RKHS norm is $\|h\|_K^2 = h^T K^{-1} h$. Let $G \sim N(0, K)$ and fix a margin $a > 0$, independent of d. The prior is first conditioned on

$$\mathcal{A}_0 = \{|G(x)| > a \text{ for every } x \in \mathcal{X}_d\}.$$

It is then conditioned on one noiseless label $y \in \{-1, 1\}$ at an arbitrary point z, that is, on $yG(z) > 0$. Under $\mathcal{A}_0$ this is equivalent to $yG(z) > a$. Write $m_G = \mathbb{E}[G \mid \mathcal{A}_0, yG(z) > a]$.

It is useful to split off independent noise:

$$G = H + \sqrt{v}\, \xi, \qquad H \sim N(0, C), \quad C = K - vI, \qquad \xi \sim N(0, I),$$

with independence. The possibly singular covariance C is interpreted on its range; let m_H be the conditional mean of H under the same event, and put $\|h\|_C^2 = h^T C^+ h$ on that range.

Theorem 1. *For every fixed $L \geq 2$ and $a > 0$, uniformly in the observed point and its label,*

$$\|m_G\|_K^2 \geq (1 - o(1))vd \log 2, \qquad \|m_H\|_C^2 \geq (1 - o(1))vd \log 2 \qquad (d \longrightarrow \infty).$$

For every fixed $\alpha \in (0, 1/2)$,

$$v \geq c_L (d \log d)^{-\beta}, \qquad \tau_{\lfloor \alpha d \rfloor} \geq c_{L,\alpha} d^{-\beta}.$$

Therefore, if $L \geq 3$, both

$$\tau_{\lfloor \alpha d \rfloor} \|m_G\|_K^2 \longrightarrow \infty, \qquad \tau_{\lfloor \alpha d \rfloor} \|m_H\|_C^2 \longrightarrow \infty.$$

In particular, no constant independent of d can bound either mean energy by $Cn/\tau_{\lfloor \alpha d \rfloor}$ uniformly in the sample size, since it fails at $n = 1$.

This theorem concerns the actual posterior after global margin conditioning. It does not exclude a bound with an additive dimension-dependent baseline, an amplitude factor, or a restriction to much larger sample sizes. The divergence conclusion is not claimed at $L = 2$. The same one-observation result applies when the observed label is generated by parity. It does not itself establish a lower or upper bound for confidently predicting parity at unobserved inputs.

2 Exact global-margin conditioning

Write Φ and ϕ for the standard normal distribution and density, and $Q = 1 - \Phi$. Define

$$B_a(t) = \Phi\left(\frac{t-a}{\sqrt{v}}\right) + \Phi\left(\frac{-t-a}{\sqrt{v}}\right), \qquad p_a(t) = \frac{\Phi((t-a)/\sqrt{v})}{B_a(t)}.$$

Conditioning on H, the independent noises give

$$d\mathcal{L}(H \mid \mathcal{A}_0) = \frac{\prod_x B_a(H(x))}{\mathbb{P}(\mathcal{A}_0)}\, dN(0, C)(H). \tag{1}$$

Moreover $\mathbb{P}(G(x) > a \mid H, \mathcal{A}_0) = p_a(H(x))$. The function B_a is even and increasing on $[0, \infty)$, as follows by differentiating. Also $p_a(-t) = 1 - p_a(t)$ and $p_a(t) \geq 1/2$ for $t \geq 0$.

Lemma 2 (A denominator from a constant shift)**.** *Set $t_0 = \sqrt{2 \log(4M)}$ and $R = a + t_0 + 1$. Then*

$$\mathbb{P}(\mathcal{A}_0) \geq \frac{1}{2} \exp\left(-\frac{R^2}{2\lambda_0}\right).$$

Proof. The event $B = \{\max_x |G(x)| \leq t_0\}$ has probability at least $1/2$ by the Gaussian tail bound and a union bound. The constant vector $h = R\mathbf{1}$ has energy R^2/λ_0, and $h + B \subset \mathcal{A}_0$. Translating the Gaussian density gives

$$\mathbb{P}(G \in h + B) = e^{-\|h\|_K^2/2} \mathbb{E}[\mathbf{1}_B(G) e^{-h^T K^{-1} G}] \geq e^{-\|h\|_K^2/2} \mathbb{P}(B).$$

The last inequality uses the symmetry of B and $\cosh u \geq 1$. □

Lemma 3 (Latent magnitudes forced by all-input conditioning)**.** *Suppose $D \geq 1$, $D < M$, and $-\log \mathbb{P}(\mathcal{A}_0) \leq D$. Set*

$$t = \sqrt{\log(M/D)}, \quad b = a + \sqrt{v}\, t, \quad c_* = \frac{e^{-3/2}}{\sqrt{2\pi}}.$$

If $e^{-2a^2/v} \leq 1/2$, then, for every fixed $z \in \mathcal{X}_d$,

$$\mathbb{P}(|H(z)| \leq b \mid \mathcal{A}_0) \leq \frac{2}{c_*} \frac{(t+1)(D+1)}{\sqrt{MD}}. \tag{2}$$

Proof. Let $q_b(H) = M^{-1}\#\{x : |H(x)| \le b\}$ and $c_b = -\log B_a(b) > 0$. On $\{q_b \ge u\}$, the density numerator in (1) is at most $e^{-Mc_b u}$. Thus

$$\mathbb{P}(q_b \ge u \mid \mathcal{A}_0) \le \min\{1, e^{D - Mc_b u}\}, \qquad \mathbb{E}[q_b \mid \mathcal{A}_0] \le \frac{D+1}{Mc_b}.$$

The expectation bound follows by integrating the tail on $u \ge 0$. For $c \ge 0$ and $t \ge 0$, shifting the Gaussian integral gives $Q(t+c) \le e^{-tc - c^2/2} Q(t)$. Therefore

$$1 - B_a(b) = Q(t) - Q(t + 2a/\sqrt{v}) \ge Q(t)/2.$$

Integrating ϕ from t to $t + (t+1)^{-1}$ yields $Q(t) \ge c_* e^{-t^2/2}/(t+1)$. Since $-\log B \ge 1 - B$,

$$Mc_b \ge \frac{c_*}{2} \frac{\sqrt{MD}}{t+1}.$$

Finally K, C, and $\mathcal{A}_0$ are invariant under cube translations, which act transitively. Consequently the probability on the left of (2) equals $\mathbb{E}[q_b \mid \mathcal{A}_0]$. □

Lemma 4 (One label reveals the large latent amplitude)**.** *Suppose $b \ge a$, and set $q = \mathbb{P}(|H(z)| \le b \mid \mathcal{A}_0)$. For the positive observed label,*

$$m_H(z) \ge b\,[1 - 2p_a(-b)](1 - q), \qquad m_G(z) \ge m_H(z).$$

The corresponding absolute-value bounds hold for a negative label. Also

$$p_a(-b) \le \exp\left(-\frac{(a+b)^2}{2v}\right).$$

Proof. Global sign symmetry gives probability $1/2$ to each label, and $\mathbb{E}[H(z) \mid \mathcal{A}_0] = 0$. Hence

$$m_H(z) = \mathbb{E}[H(z) \operatorname{sign} G(z) \mid \mathcal{A}_0] = \mathbb{E}[|H(z)|(2p_a(|H(z)|) - 1) \mid \mathcal{A}_0].$$

The integrand is nonnegative and increasing in $|H(z)|$. To see the needed monotonicity, the odds $p_a(u)/(1 - p_a(u)) = \Phi((u-a)/\sqrt{v})/\Phi((-u-a)/\sqrt{v})$ increase with u. Restricting to $|H(z)| > b$ proves the first bound. For $u \ge 0$, the normal Mills ratio $\phi(u)/Q(u)$ is at least u, because it is the conditional mean of a standard normal given that it exceeds u. Integrating gives $Q(u) \le \frac{1}{2} e^{-u^2/2}$. Since $b \ge a$,

$$p_a(-b) \le \frac{Q((a+b)/\sqrt{v})}{\Phi((b-a)/\sqrt{v})} \le e^{-(a+b)^2/(2v)}.$$

Finally, given H, $\mathcal{A}_0$, and $G(z) > a$, the residual noise at z is a one-sided truncated normal with positive mean. Thus $\mathbb{E}[G(z) \mid H, \mathcal{A}_0, G(z) > a] \ge H(z)$, which implies $m_G(z) \ge m_H(z)$. Global sign symmetry handles the negative label. □

3 Elementary spectral estimates

The estimates needed here follow without a sharp eigenvalue asymptotic.

Lemma 5 (Coefficient tails)**.** *Write $F(u) = \sum_{m \ge 0} c_m u^m$. Then $c_m \ge 0$, $\sum_m c_m = 1$, $c_1 = (2/\pi)^{L-1} > 0$, and*

$$c_L R^{-\beta} \le \sum_{m \ge R} c_m \le C_L R^{-\beta} \qquad (R \ge 2).$$

Proof. The nonnegative power series of arcsine and its compositions gives the first statements. The expansion $A(1-s) = 1 - (2\sqrt{2}/\pi)s^{1/2} + O(s^{3/2})$ implies by induction that $1 - F(1-s) \asymp_L s^\beta$ as $s \downarrow 0$. Let $T(R) = \sum_{m \geq R} c_m$. Since

$$1 - F(1 - 1/R) \geq [1 - (1 - 1/R)^R]T(R),$$

the upper bound follows. Consequently $\sum_{m<R} mc_m \leq \sum_{j<R} T(j) \leq C'_L R^{1-\beta}$. For any fixed small $\epsilon > 0$, Bernoulli's inequality gives

$$c_L \epsilon^\beta R^{-\beta} \leq 1 - F(1 - \epsilon/R) \leq C'_L \epsilon R^{-\beta} + T(R).$$

Choosing ϵ sufficiently small proves the lower bound, since $\beta < 1$. □

Lemma 6 (Bounds for the constant mode, smallest mode, and tail)**.** *For fixed $\alpha \in (0, 1/2)$ and sufficiently large d,*

$$\lambda_0 \geq \frac{(2/\pi)^{L-1}}{d+1}, \quad c_L(d \log d)^{-\beta} \leq v \leq C_L d^{-\beta}, \quad c_{L,\alpha} d^{-\beta} \leq \tau_{\lfloor \alpha d \rfloor} \leq C_{L,\alpha} d^{-\beta}.$$

Proof. Put $D = d + 1$. The degree-m kernel $[(1 + x \cdot y)/D]^m$ has normalized Walsh eigenvalues equal to the law of a walk on subsets of $[d]$: at each step choose uniformly from $\{0, 1, \ldots, d\}$; choice zero leaves the subset fixed, and choice $i \geq 1$ toggles i. Write S_m for the resulting subset. Thus

$$\lambda_j = \sum_m c_m \mathbb{P}(S_m = S) \quad (|S| = j), \qquad \tau_r = \sum_m c_m \mathbb{P}(|S_m| > r).$$

The $m = 1$ term proves the constant-mode bound. Since $|S_m| \leq m$, Lemma 5 proves the upper tail bound. At least $M/2$ characters have degree greater than $\lfloor d/4 \rfloor$ for sufficiently large d. Since every matrix eigenvalue is at least v, their normalized trace is at least $v/2$; the upper tail bound therefore gives $v \leq C_L d^{-\beta}$.

For a lower bound on every eigenvalue, Fourier inversion for the walk gives

$$|M\mathbb{P}(S_m = S) - 1| \leq \sum_{j=1}^{d} \binom{d}{j} |1 - 2j/D|^m \leq (1 + e^{-2m/D})^D - 1.$$

For the second inequality, pair j with $D - j$ and use $\binom{d}{j} + \binom{d}{j-1} = \binom{D}{j}$ and $|1 - 2j/D| \leq e^{-2\min(j, D-j)/D}$. If $m \geq D \log(4D)$, the final expression is less than $1/2$. It follows that every λ_j is at least $\frac{1}{2M} \sum_{m \geq D \log(4D)} c_m$. Lemma 5 proves the lower bound on v.

Finally

$$\mathbb{E}|S_m| = \frac{d}{2} \left[1 - (1 - 2/D)^m\right].$$

Choose a constant C_α so that for every $m \geq C_\alpha D$ this expectation is at least $(\alpha + \eta)d$ for some $\eta > 0$. Since $|S_m| \leq d$, it follows that $\mathbb{P}(|S_m| > \lfloor \alpha d \rfloor) \geq \eta/(1 - \alpha)$. Summing over these m and applying Lemma 5 proves the lower tail bound. □

4 Proof of the theorem and a distinction for posterior draws

Proof of Theorem 1. Take

$$D = 1 + \log 2 + \frac{(a + \sqrt{2 \log(4M)} + 1)^2}{\lambda_0}.$$

Lemmas 2 and 6 give an admissible bound $D = O_{L,a}(d^2)$ on $-\log \mathbb{P}(\mathcal{A}_0)$. Since $v \to 0$, Lemma 3 applies for large d. With $b = a + \sqrt{v \log(M/D)}$, it gives $q = o(1)$ in Lemma 4. Also $p_a(-b) = o(1)$, so

$$|m_H(z)| \geq (1 - o(1))b, \qquad |m_G(z)| \geq (1 - o(1))b.$$

The RKHS evaluation inequality, valid also for a singular covariance on its range, gives

$$\|m_H\|_C^2 \geq \frac{m_H(z)^2}{C(z,z)} = \frac{m_H(z)^2}{1 - v}, \qquad \|m_G\|_K^2 \geq \frac{m_G(z)^2}{K(z,z)} = m_G(z)^2.$$

Because $\log(M/D) = d \log 2 - O(\log d)$, these prove the energy lower bounds. The remaining estimates follow from Lemma 6. In particular

$$vd\, \tau_{\lfloor \alpha d \rfloor} \geq c_{L,\alpha} \frac{d^{1-2\beta}}{(\log d)^{\beta}} \longrightarrow \infty \qquad (\beta < 1/2).$$

□

A proposed bound for a posterior *draw* has a further, elementary obstruction. For any positive definite $M \times M$ covariance, any positive global margin, and any compatible noiseless training signs,

$$\mathbb{E}[\|G\|_K^2 \mid \text{global margins and training signs}] \geq M. \tag{3}$$

Indeed write the whitened Gaussian as $K^{-1/2}G = RU$, where the radius R and direction U are independent and $\mathbb{E}R^2 = M$. For each direction satisfying the signs, the global margins impose only a lower cutoff on R. Truncating a nonnegative random variable above a lower cutoff increases its mean square. Averaging over the conditional directions proves (3). Thus draw-energy estimates must specify a dimension baseline or a centered or projected quantity.

The obstruction proved here concerns amplitude and energy. Predictive correctness with probability strictly greater than $2/3$ is a separate quantity: a large posterior mean at the observed input does not imply confident correct prediction on unobserved inputs. No sample-complexity conclusion is inferred from this note.